

Day 10 — Summative + Share

Module: Magic Squares (*Bhadragaṇita*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will (a) complete the summative quiz, and (b) present their original magic-square project to the class.

Materials

- Printed summative quiz (one per student) — `quizzes/summative.md`
- Project artefacts (posters, devices for videos, written exposition)
- Stopwatch (for 2-minute presentation timer)

Lesson flow

Summative quiz (15 min)

- Closed notebook. Students complete `quizzes/summative.md` individually.
- 15 min total: 3 min reading, 12 min solving.

Presentations (25 min)

- Each pair gets **2 minutes** to present + 1 minute Q&A.
- With ~10–12 pairs in a typical class, this fits in 25 min.
- Order: teacher draws pair names randomly.

What each presentation must show:

- The constructed magic square (size, magic constant verified).
- Citation of the source (Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, 1356 CE).
- One sentence on *why this particular design* — what choice did the pair make?

Wrap-up (5 min)

- One question per student on a slip of paper: “*What is one thing you’ll remember about magic squares one year from now?*”
- Collect. Read at random — 2–3 aloud.
- Close with: “*You all now know an algorithm older than most countries’ modern names. Use it. Show someone.*”

Student-facing content

Today is the share.

Bring: - Your project artefact (poster / video / written exposition). - Your two-minute presentation prepared. - Your full attention for your classmates’ presentations — you will give feedback.

Reminder: **Verify before you present.** Sum at least 3 rows, 3 columns, and both diagonals out loud (or in your head). If they don't match the magic constant, fix the error *before* you stand up.

Homework — none

The module ends here. The project artefact stays with the school until the end of term as part of the module's portfolio. Reflection slips are collected.

Differentiation

- **One tier up:** Pairs who finish early help check others' projects against the rubric (specifically, the “method demonstrated correctly” criterion).
- **One tier down:** A pair that didn't quite finish presents a smaller artefact (e.g. 3×3 instead of 5×5). They still meet the citation and verification requirements.

Teacher notes

- Keep the presentations tight. 2 minutes is short on purpose — encourages preparation.
- The Q&A minute is for *one* clarifying question, not a defence. Cap it strictly.
- Score against `assessments/rubric.md` during the share — don't try to score after. Live scoring keeps things fast.
- For students who freeze: have a partner take over. The presentation is paired; both partners speak counts towards the rubric.

Citation(s) used in this lesson

- Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī*, Book 14 (*Bhadragaṇita*), 1356 CE — referenced in project artefacts.